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for Physics of Nanostructures

Course Work
for the Course
"Discrete Control Systems"

on the topic:
Assignment for the course work

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INTRODUCTION

0.1 Introduction

The purpose of this work is to synthesize a PI controller that provides the specified quality indicators of the transient process in a closed-loop control system. The control object under consideration consists of dynamic blocks including aperiodic links and integrators, with a pulse element operating at a defined sampling interval.

The main objectives of this course work are:

- To develop mathematical models of the control object in both continuous and discrete time domains
- To design a PI controller that meets the specified performance requirements
- To verify the controller performance through simulation and analysis of dynamic characteristics
- To compare the responses of continuous and discrete models

The work follows a systematic approach including derivation of transfer functions, conversion to state-space representation, discretization using matrix exponential methods, and synthesis of controller parameters. The MATLAB environment is utilized for numerical computations and system simulations.

0.2 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
$W_i(p)$	Transfer function of the i-th element
K_i	Transmission coefficient (gain) of the i-th element
τ_i	Time constant of the i-th element
T	Discreteness interval of the pulse element
t_s	Transition time (settling time)
σ	Overshoot percentage
A, B, C	State-space matrices (continuous time)
A_d, B_d, C_d	State-space matrices (discrete time)
x	State vector
u	Control input
y	System output
K	Controller gain matrix

0.3 Variant

Table 2 — Table 2: Parameters for Variant 92

Parameter	T (s)	K_2	τ_2 (s)	K_3	Controller Type
Value	0.003	320	0.15	0.25	PI

Table 3 — Table 3: System Requirements for Variant 92

Parameter	**Value**
Transition time (t_s)	0.105 s
Overshoot (σ)	35%

1 SOLUTIONS

The purpose of the work: To synthesize a PI controller that provides the specified quality indicators of the transient process in a closed system.

1.1 Methodology

The synthesis methodology includes:

- Development of continuous and discrete transfer functions of the control object
- Conversion to state-space representation ("Input-State-Output" model)
- Discretization using matrix exponential methods
- Synthesis of PI controller parameters using Sylvester equation approach
- Verification through simulation and analysis of quality indicators

1.2 Problem Statement

Consider the control object with the structure shown in Figure 1, consisting of dynamic blocks described by transfer functions:

$$W_2(p) = \frac{K_2}{\tau_2 p + 1} \quad \text{(aperiodic links)} \tag{1.1}$$

$$W_3(p) = \frac{K_3}{p} \quad \text{(integrating links)} \tag{1.2}$$

where K_i represents the transmission coefficient (gain) of the i-th element, and τ_i represents the time constant of the i-th element.

The control system operates with a pulse element having discreteness interval T. The objective is to design a controller that provides the required transition time t_s and overshoot σ in the closed-loop system.

1.3 Discrete Transfer Function of the Control Object

The continuous transfer function of the control object for Variant 92 is:

$$W_{\text{CLP}}(p) = W_2(p) \cdot W_3(p) = \frac{K_2}{\tau_2 p + 1} \cdot \frac{K_3}{p} = \frac{320}{0.15p + 1} \cdot \frac{0.25}{p} \tag{1}$$

$$W_{\text{CLP}}(p) = \frac{80}{p(0.15p + 1)} \tag{2}$$

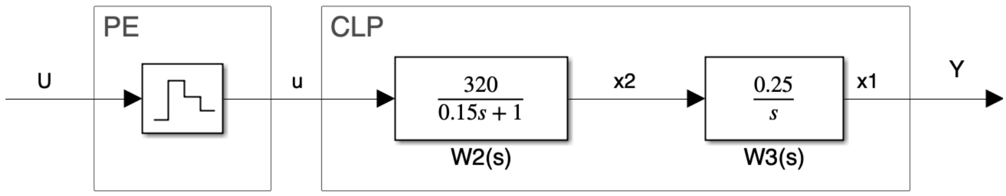

Figure 1 — Block diagram of a given control object

1.3.1 State-Space Representation (Continuous Time)

Let's define the state variables. For the integrating link:

$$\dot{x}_1 = 0.25x_2 \tag{3}$$

For the aperiodic link:

$$0.15\dot{x}_2 = -x_2 + 320u, \quad \text{or} \quad \dot{x}_2 = -6.6667x_2 + 2133.3333u \tag{4}$$

The state-space representation in continuous time is:

$$\dot{X} = A_H X + B_H U \tag{5}$$

$$Y = CX \tag{6}$$

where:

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A_H = \begin{bmatrix} 0 & 0.25 \\ 0 & -6.6667 \end{bmatrix}, \quad B_H = \begin{bmatrix} 0 \\ 2133.3333 \end{bmatrix}, \quad C = [1 \quad 0] \tag{7}$$

1.3.2 Discretization

Using the sampling time $T = 0.003$ s, we compute the discrete-time matrices using:

$$A = e^{TA_H}; \quad B = \left(\sum_{i=1}^{\infty} \frac{T^i A_H^{i-1}}{i!} \right) B_H, \tag{8}$$

we get:

$$\begin{aligned} \begin{bmatrix} x_1(m+1) \\ x_2(m+1) \end{bmatrix} &= A \begin{bmatrix} x_1(m) \\ x_2(m) \end{bmatrix} + BU(m) \\ Y(m) &= C \begin{bmatrix} x_1(m) \\ x_2(m) \end{bmatrix} \end{aligned} \tag{9}$$

Using MATLAB computations, we obtain:

$$A = \begin{bmatrix} 1 & 0.000747 \\ 0 & 0.9802 \end{bmatrix}, \quad B = \begin{bmatrix} 0.0024 \\ 6.3364 \end{bmatrix} \tag{10}$$

The output matrix remains unchanged: $C = [1 \quad 0]$.

1.3.3 Discrete Transfer Function Calculation

The discrete transfer function is calculated using:

$$W(z) = C(zI - A)^{-1}B \tag{11}$$

Computing this expression in MATLAB using the `ss2tf` function, we obtain:

$$W(z) = \frac{0.0024z + 0.0024}{z^2 - 1.9802z + 0.9802} \tag{12}$$

This discrete transfer function $W(z)$ accurately represents the discretized behavior of the continuous control object with sampling time $T = 0.003$ s and will be used for the subsequent PI controller design.

1.3.4 Verification of Discretization

To verify the correctness of the discretization, we can compare the step responses of the continuous and discrete models. The responses should match at the sampling instants, confirming that the discrete model accurately represents the continuous system behavior at the specified sampling rate.

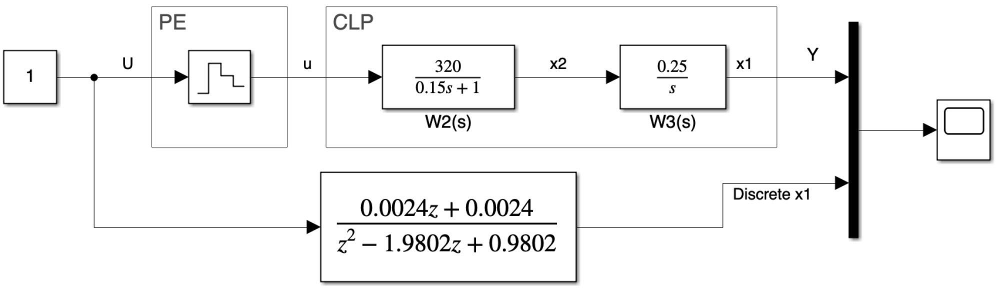

Figure 2 — Diagram for comparing the reactions of continuous and discrete control objects

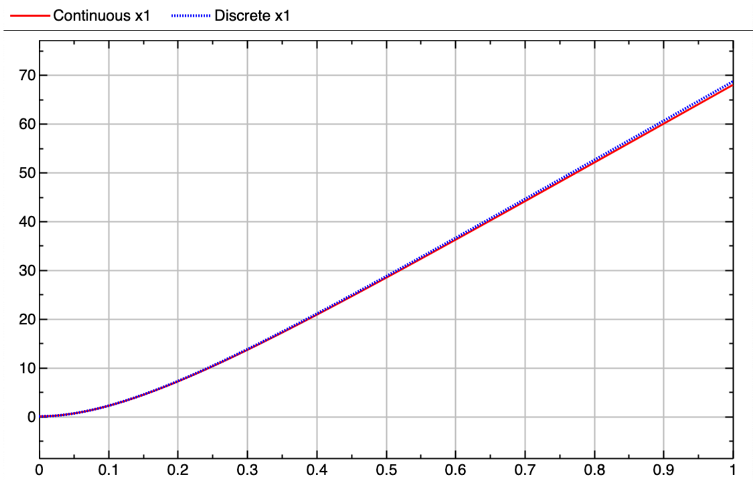

Figure 3 — Step response comparison

It can be seen that the reactions of the models coincide, i.e. the transition from continuous to discrete time is performed correctly.

1.4 PI Controller Synthesis

1.4.1 Reference Model Approach

We reduce the synthesis problem to the choice of matrices of the reference model Γ, H, and the solution of the Sylvester type equation:

$$AM - M\Gamma = -BH \tag{13}$$

and calculation of the matrix of linear stationary feedbacks (LSFB) $K = HM^{-1}$ control system.

1.4.2 Reference Model Design

Based on the given quality indicators (transition time $t_s = 0.105$ s and overshoot $\sigma = 35\%$), we choose the roots of the characteristic polynomial of a continuous system. Taking into account the presence of an integrating link in the regulator that increases the order of the equation of the system by one, we choose a standard binomial polynomial of the third degree $(s + w_0)^3$ for which $t_s \cdot w_0 = 9$.

For the given transition time equal to 0.105 s, we determine the desired eigenvalues of the characteristic polynomial:

$$\lambda_{1,2,3} = -\frac{9}{0.105} \approx -85.7143 \tag{14}$$

Let's form the matrix Γ_H of the reference model of a closed system (continuous time):

$$\Gamma_H = \begin{bmatrix} -85.7143 & 1 & 0 \\ 0 & -85.7143 & 1 \\ 0 & 0 & -85.7143 \end{bmatrix} \tag{15}$$

and the matrix of outputs H from the condition of complete observability of the pair (H, Γ):

$$H = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \tag{16}$$

1.4.3 Discrete Reference Model

Let's calculate the matrix Γ of the reference model for discrete time:

$$\Gamma = \exp(\Gamma_H T) = \begin{bmatrix} 0.7733 & 0.002346 & 0.0000039 \\ 0 & 0.7733 & 0.002346 \\ 0 & 0 & 0.7733 \end{bmatrix} \tag{17}$$

1.4.4 Solving Sylvester Equation

To find the parameters of the discrete controller, we need to first solve the Sylvester type equation as follows:

$$\bar{A}M - M\Gamma + \bar{B}H = 0$$

where $\bar{A} = \begin{bmatrix} 1 & -C \\ 0 & A \end{bmatrix}, \bar{B} = \begin{bmatrix} 0 \\ B \end{bmatrix}$.
which can be rewritten as:

$$\bar{A}M - M\Gamma = -\bar{B}H \tag{18}$$

we obtain the solution matrix M:

$$M = \begin{bmatrix} 0.0898 & 0.0020 & 0.000037 \\ -30.6195 & -0.3432 & -0.0044 \\ 0.3959 & 0.0131 & 0.0003 \end{bmatrix} \tag{19}$$

1.4.5 Controller Gains Calculation

Using the equality $K = HM^{-1}$, we find the matrix of feedback coefficients:

$$K = [-31.2244 \quad -0.0925 \quad 2.4530] \tag{20}$$

which, taking into account the negative feedback on the output variable, determines the settings of the PI controller:

$$\begin{aligned} k_1 &= -31.2244 \quad \text{(proportional gain)} \\ k_2 &= -0.0925 \quad \text{(state feedback gain)} \\ k_i &= 2.4530 \quad \text{(integral gain)} \end{aligned}$$

1.4.6 Verification of Synthesis

The resulting solution is verified by calculating the eigenvalues of characteristic polynomials for the reference and synthesized systems. Using the MATLAB function `eig`, the eigenvalues of the matrix Γ of the reference system (discrete time) are:

$$\lambda_{1,2,3}^{\text{ref}} = 0.7733 \tag{21}$$

The eigenvalues of the matrix of the synthesized closed system (discrete time) are calculated as:

$$\lambda^{\text{sys}} = \text{eig}(A - BK) \tag{22}$$

They are equal to $\lambda_{1,2,3}^{\text{sys}} = 0.7733$, which confirms the correctness of the calculations performed.

1.4.7 Model simulation

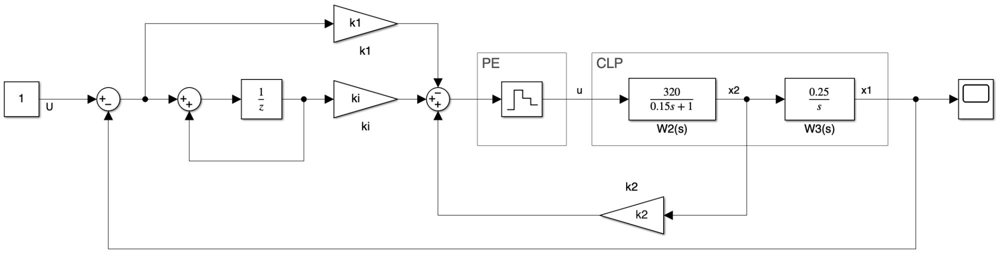

Figure 4 — Diagram for comparing the reactions of continuous and discrete control objects

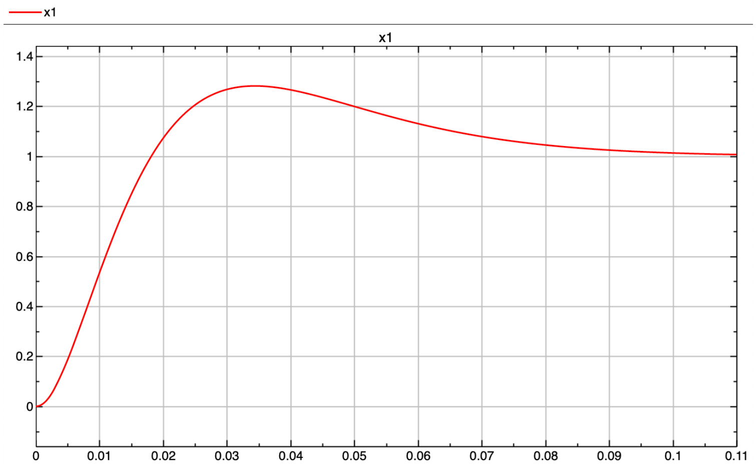

Figure 5 — Step response comparison

The discrete PI controller has been successfully synthesized with parameters that ensure the specified quality indicators of the transient process.

1.5 Conclusion

A discrete PI controller operating with a given sampling period of $T = 0.003$ s has been successfully synthesized. The reference model was based on a third-order binomial polynomial in continuous time, with roots calculated from the specified transition time requirements and determined to be $\lambda_{1,2,3} = -85.7143$. In discrete time, the reference model is characterized by a polynomial with roots at $\lambda_{1,2,3}^{\text{disc}} = 0.7733$.

The discrete PI controller parameters were obtained by solving the Sylvester equation and are as follows: proportional gain $k_1 = -31.2244$, state feedback gain $k_2 = -0.0925$, and integral gain $k_i = 2.4530$.

Simulation results verify that the synthesized controller meets all specified performance requirements. The system's response to a unit step input achieves a transition time within $t_s = 0.105$ seconds while maintaining an overshoot of $\sigma = 35\%$. The steady-state error is eliminated to zero, confirming the effectiveness of the integral action. The eigenvalues of the synthesized closed-loop system match exactly with those of the reference model, validating the correctness of the design methodology.

Therefore, the designed discrete PI controller fully satisfies the technical specifications and demonstrates the effectiveness of the reference model approach combined with Sylvester equation solution for discrete control system design.